Weil Conjectures

Section 1: Introduction

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1 Introduction

These notes prove the Weil conjectures, which concern the number of solutions of polynomial equations over finite fields. Weil saw that rationality and the functional equation would follow from a cohomology theory for varieties over finite fields with a Lefschetz fixed point formula and Poincaré duality, while the deepest conjecture, an analogue of the Riemann hypothesis, needs in addition a bound on the eigenvalues of Frobenius. Following this plan, we construct such a theory, ℓ-adic cohomology, and prove the trace formula and Poincaré duality for it. Then we prove the bound, which is the Deligne–Laumon theorem.

1.1 The Weil conjectures

Let X_0 be a smooth projective geometrically connected curve of genus g over the finite field $\mathbf{F}_q$, and let N_m be the number of its points with coordinates in $\mathbf{F}_{q^m}$. The projective line has exactly $1 + q^m$ such points, and the *Hasse–Weil bound* says that every curve of genus g comes close to this count:

$$|N_m - (1 + q^m)| \leq 2g\, q^{m/2}.$$

Hasse proved it for elliptic curves [Has36] and Weil for all curves [Wei48a].

To state Weil's conjectures in higher dimension we put all the numbers N_m into one generating function. For a separated scheme X_0 of finite type over $\mathbf{F}_q$, its *zeta function* is

$$Z(X_0, T) = \exp\Big(\sum_{m\geq 1} \#X_0(\mathbf{F}_{q^m})\frac{T^m}{m}\Big) \in 1 + T\mathbf{Q}[[T]].$$

A closed point x of X_0 with residue field $\mathbf{F}_{q^{\deg x}}$ accounts for exactly $\deg x$ points of $X_0(\mathbf{F}_{q^m})$ when $\deg x$ divides m, and for none otherwise. Hence $\#X_0(\mathbf{F}_{q^m}) = \sum_{\deg x | m} \deg x$, so

$$\sum_{m\geq 1} \#X_0(\mathbf{F}_{q^m})\frac{T^m}{m} = \sum_{x\in|X_0|}\sum_{k\geq 1}\frac{T^{k\deg x}}{k} = -\sum_{x\in|X_0|}\log(1-T^{\deg x}).$$

Only finitely many closed points have a given degree, so exponentiating gives the Euler product

$$Z(X_0, T) = \prod_{x\in|X_0|}\big(1-T^{\deg x}\big)^{-1},$$

whose factors are power series with integer coefficients. So the coefficients of $Z(X_0, T)$ are integers. For example, $\#\mathbf{P}^1(\mathbf{F}_{q^m}) = 1 + q^m$ gives $Z(\mathbf{P}^1, T) = 1/((1-T)(1-qT))$. Set $T = q^{-s}$. Since the residue field of x has $N(x) = q^{\deg x}$ elements, the Euler product becomes

$$\zeta_{X_0}(s) = Z(X_0, q^{-s}) = \prod_{x\in|X_0|}\big(1-N(x)^{-s}\big)^{-1},$$

the analogue of Riemann's

$$\zeta(s) = \sum_{n\geq 1} n^{-s} = \prod_p (1-p^{-s})^{-1},$$

in which the closed points of $\operatorname{Spec}\mathbf{Z}$ have residue fields with p elements. Riemann conjectured that the nontrivial zeros of $\zeta(s)$ lie on the line $\operatorname{Re} s = \frac{1}{2}$. Weil conjectured the following.

1. *Rationality.* $Z(X_0, T)$ is a rational function of T.
2. *Functional equation.* If X_0 is smooth and proper of pure dimension d, then $Z(X_0, 1/(q^d T)) = \pm q^{d\chi/2}T^\chi Z(X_0, T)$, where $\chi = \sum_i (-1)^i b_i$ is the Euler characteristic, with b_i as in (4). For a curve of genus g it equals $2-2g$.
3. *Riemann hypothesis.* If X_0 is smooth and proper of pure dimension d, then

$$Z(X_0, T) = \frac{P_1(T)P_3(T)\cdots P_{2d-1}(T)}{P_0(T)P_2(T)\cdots P_{2d}(T)},$$

   where $P_i \in \mathbf{Z}[T]$ has constant term 1, and its reciprocal roots (the numbers α with $P_i(1/\alpha) = 0$) are algebraic integers all of whose complex conjugates have absolute value $q^{i/2}$.
4. *Betti numbers.* The degree b_i of P_i behaves like a Betti number. For instance, $b_i = b_{2d-i}$, as for Poincaré duality on a compact manifold of real dimension $2d$. More precisely, suppose that X_0 comes from characteristic zero: there is a smooth projective variety Y over a number field K and a prime $\mathfrak{p}$ of K with residue field $\mathbf{F}_q$ such that Y extends to a smooth projective scheme over the local ring $\mathcal{O}_{K,\mathfrak{p}}$ whose special fibre is X_0. Then b_i is the ith Betti number of the compact complex manifold $Y(\mathbf{C})$, for any embedding $K \subset \mathbf{C}$. For a curve of genus g this gives $b_0 = b_2 = 1$ and $b_1 = 2g$.

For a curve the conjectures recover the Hasse–Weil bound. Here $Z(X_0,T) = P_1(T)/((1-T)(1-qT))$ with P_1 of degree $2g$, and statement (3) says that $P_1(T) = \prod_{j=1}^{2g}(1-\alpha_j T)$ with $|\alpha_j| = q^{1/2}$. Comparing the logarithmic derivatives of both sides with the definition of Z gives

$$N_m = 1 + q^m - \sum_{j=1}^{2g} \alpha_j^m,$$

and since each $|\alpha_j^m| = q^{m/2}$, the Hasse–Weil bound follows. This is also why (3) is called the Riemann hypothesis. In the variable s with $T = q^{-s}$, the zeros of $P_1(q^{-s})$ satisfy $|q^{-s}| = q^{-1/2}$, that is, they lie on the line $\operatorname{Re} s = \frac{1}{2}$, exactly where Riemann conjectured the zeros of $\zeta(s)$ to lie.

We follow one example through the notes, the elliptic curve

$$C_0 \colon\ y^2 = x^3 - x \qquad \text{over } \mathbf{F}_3.$$

Counting points, including the point at infinity, gives $N_1 = 4$, $N_2 = 16$ and $N_3 = 28$. Suppose the Weil conjectures hold for C_0. Then $Z(C_0,T) = P_1(T)/((1-T)(1-3T))$, where P_1 has degree $b_1 = 2g = 2$. Its leading coefficient is $q = 3$, because by duality it is the determinant of Frobenius on H^1 (Example 10.23*). So $P_1 = 1 - aT + qT^2$, and comparing the coefficients of T in Z gives $a = 1 + q - N_1 = 0$. Hence $P_1(T) = 1 + 3T^2$, whose reciprocal roots $\pm i\sqrt{3}$ have absolute value $3^{1/2}$. Statements (1)–(4) alone would not fix the leading coefficient: $1 - 3T^2$ also satisfies them, with the sign $-$ in (2), and it is the count $N_2 = 16$ that excludes it. As a check, the formula $N_m = 1 + 3^m - \big((i\sqrt{3})^m + (-i\sqrt{3})^m\big)$ gives 16 and 28 for $m = 2, 3$. We return to this curve after duality, after the trace formula and at the end (Examples 10.23*, Example 12.32* and Example 19.9*).

1.2 Weil cohomology

Write $X = X_0 \times_{\mathbf{F}_q} \overline{\mathbf{F}}_q$, and let $F : X \to X$ be the *Frobenius*: if X_0 is cut out of an affine space by equations with coefficients in $\mathbf{F}_q$, then F raises each coordinate to its q-th power. A point of X is fixed by F^m exactly when its coordinates lie in $\mathbf{F}_{q^m}$, so $\#X_0(\mathbf{F}_{q^m})$ is the number of fixed points of F^m. Weil's idea was to count these fixed points as in topology.

Suppose that we had a cohomology theory with compact supports, assigning to X finite-dimensional vector spaces $H_c^i(X)$ over a field of characteristic zero on which F acts. Compact supports are needed because the trace formula below has to hold for every X_0, proper or not: point counts are additive when X_0 is cut into an open subscheme and its complement, and compactly supported cohomology has the matching long exact sequence. For proper X_0 we write $H^i(X) = H_c^i(X)$; for the conjectures, which concern smooth proper varieties, this is the only cohomology we need. Suppose that the theory satisfies the following.

- (a) *Finiteness.* $H_c^i(X) = 0$ unless $0 \le i \le 2\dim X$.
- (b) *Trace formula.* The fixed points of F^m are counted by a Lefschetz formula,

$$\#X_0(\mathbf{F}_{q^m}) = \sum_i (-1)^i \operatorname{Tr}(F^m \mid H_c^i(X)).$$

- (c) *Poincaré duality.* For smooth proper X_0 of pure dimension d, cup product gives a perfect pairing between H^i and H^{2d-i}, and $\langle Fx, Fy\rangle = q^d\langle x, y\rangle$. The factor q^d is the degree of F. It is the scalar by which F acts on the one-dimensional space H^{2d} when X is connected.
- (d) *Purity.* For smooth proper X_0, the eigenvalues of F on $H^i(X)$ are algebraic integers all of whose complex conjugates have absolute value $q^{i/2}$.

Then the conjectures follow. For rationality we compare logarithms. For an endomorphism F of a finite-dimensional vector space V we have

$$\log\det(1-TF\mid V)^{-1}=\sum_{m\geq 1}\operatorname{Tr}(F^m\mid V)\frac{T^m}{m},$$

as one sees by triangulating F: if F has eigenvalues $\alpha_1,\dots,\alpha_b$, both sides equal $\sum_j\sum_m \alpha_j^m T^m/m$. Substituting the trace formula (b) into the definition of Z therefore gives

$$\log Z(X_0,T)=\sum_{m\geq 1}\sum_i(-1)^i\operatorname{Tr}\big(F^m\mid H_c^i(X)\big)\frac{T^m}{m}=\sum_i(-1)^{i+1}\log\det\big(1-TF\mid H_c^i(X)\big),$$

and so

$$Z(X_0,T)=\prod_i\det\big(1-TF\mid H_c^i(X)\big)^{(-1)^{i+1}},$$

a finite product by (a). In particular $Z(X_0,T)$ is rational, and for smooth proper X_0 the polynomial $P_i(T)$ of the conjectures is $\det(1-TF\mid H^i(X))$. By duality, the eigenvalues in degree $2d-i$ are the numbers q^d/α for the eigenvalues α in degree i, and this gives the functional equation. Purity gives the Riemann hypothesis.

Purity also shows that the factors do not depend on the cohomology theory. Indeed, the factors in different degrees have reciprocal roots of different absolute values, so no cancellation can occur in the alternating product. Each factor is therefore determined by $Z(X_0,T)$ alone, and so is its degree b_i.

Which cohomology could this be? On an irreducible scheme a constant sheaf is flasque, so its Zariski cohomology vanishes in positive degrees. It cannot see the $2g$ independent loops of a curve of genus g. Coherent cohomology sees $H^1(C,\mathcal{O}_C)$, of dimension g. But this is a vector space over $\overline{\mathbf{F}}_q$, and traces on it determine numbers of points only modulo p.

An argument of Serre (see [Den22]) shows that the coefficient field cannot be $\mathbf{Q}$, $\mathbf{R}$ or $\mathbf{Q}_p$ either. Let C be a supersingular elliptic curve over $\overline{\mathbf{F}}_p$. Then $D=\operatorname{End}(C)\otimes\mathbf{Q}$ is a quaternion algebra over $\mathbf{Q}$, and $D\otimes\mathbf{R}$ and $D\otimes\mathbf{Q}_p$ are still division algebras [Deu41]. Endomorphisms of C would act on $H^1(C)$, a vector space of dimension $2g=2$ over the coefficient field K. This gives a map of four-dimensional K-algebras $D\otimes K\to M_2(K)$. The map is injective because $D\otimes K$ is simple, hence it is an isomorphism. That is impossible if $D\otimes K$ is a division algebra. For $\ell\neq p$, on the other hand, $D\otimes\mathbf{Q}_\ell\simeq M_2(\mathbf{Q}_\ell)$, and the argument does not exclude $\mathbf{Q}_\ell$.

The theory that works is étale cohomology. Its idea is to replace the open subsets of the Zariski topology by étale maps $U\to X$, which play the role of small open neighbourhoods in topology: like a covering space, an étale map is a local isomorphism in the sense of the implicit function theorem. A sheaf in this topology assigns a group to every étale map $U\to X$, compatibly with restriction, and étale cohomology is the derived functor of global sections. Because there are many more étale maps than open subsets, a sheaf can now have nonzero cohomology in positive degrees.

For a curve we can already see the missing loops in degree one. Let C be a smooth projective connected curve of genus g over $\overline{\mathbf{F}}_q$, and let n be prime to q. Three étale sheaves are involved: $\mathbf{G}_m$, which sends $U\to C$ to the unit group $\mathcal{O}(U)^\times$; the same sheaf with the nth power map $t\mapsto t^n$; and its kernel μ_n, the sheaf of nth roots of unity. The sequence

$$0 \to \mu_n \to \mathbf{G}_m \xrightarrow{t \mapsto t^n} \mathbf{G}_m \to 0$$

is exact in the étale topology. Indeed, surjectivity of a map of sheaves only has to hold locally, and an nth root of a unit a exists on the étale cover defined by $t^n = a$, which is étale because n is invertible. (In the Zariski topology the last map is not surjective.) As in the Zariski topology, $H^1(C_{ét}, \mathbf{G}_m)$ is the Picard group of C (Hilbert's Theorem 90). The global units of C are $\overline{\mathbf{F}}_q^\times$, in which nth roots exist, so $H^0(\mathbf{G}_m) \xrightarrow{n} H^0(\mathbf{G}_m)$ is surjective. The long exact sequence therefore identifies $H^1(C_{ét}, \mu_n)$ with the kernel of multiplication by n on $H^1(C_{ét}, \mathbf{G}_m) = \mathrm{Pic}(C)$.

To compute this kernel we use the degree. It gives an exact sequence $0 \to \mathrm{Pic}^0(C) \to \mathrm{Pic}(C) \xrightarrow{\deg} \mathbf{Z} \to 0$, and since $\mathbf{Z}$ has no torsion, every line bundle killed by n has degree zero. So $\mathrm{Pic}(C)[n] = \mathrm{Pic}^0(C)[n]$. The group $\mathrm{Pic}^0(C)$ is the group of points of the *Jacobian* J of C, an abelian variety of dimension g (Section C.4*), and on an abelian variety of dimension g the n-torsion points form a group $(\mathbf{Z}/n)^{2g}$ when n is prime to the characteristic (Proposition B.14*). Altogether,

$$H^1(C_{ét}, \mu_n) \simeq \mathrm{Pic}(C)[n] = J[n] \simeq (\mathbf{Z}/n)^{2g}$$

(Proposition D.3*), which is the size of H^1 of a compact Riemann surface of genus g with coefficients $\mathbf{Z}/n$. Now let $n = \ell^k$ and let $k \to \infty$. The groups $J[\ell^k]$, with the maps $J[\ell^{k+1}] \to J[\ell^k]$ given by multiplication by ℓ, form the *Tate module* $T_\ell J = \varprojlim_k J[\ell^k]$, free of rank $2g$ over $\mathbf{Z}_\ell$ (Proposition B.14*). On the cohomology side these maps correspond to $\mu_{\ell^{k+1}} \to \mu_{\ell^k}$, $z \mapsto z^\ell$, so the identification above passes to the limit:

$$\varprojlim_k H^1(C_{ét}, \mu_{\ell^k}) \simeq T_\ell J$$

(Proposition C.17*). In other words, the Tate module is the first cohomology of C with coefficients in the inverse system $(\mu_{\ell^k})_k$, which is written $\mathbf{Z}_\ell(1)$ (we come back to what this limit means below). Weil used $T_\ell J$ in his proof for curves, in place of an H^1 that was not yet available.

Frobenius acts on both sides. Let π be the endomorphism of J induced by Frobenius; it raises the coordinates of torsion points to the q-th power, and under the isomorphism above it is the action of the arithmetic Frobenius $\sigma : z \mapsto z^q$ in $\mathrm{Gal}(\overline{\mathbf{F}}_q/\mathbf{F}_q)$ on the torsion points. The map F on cohomology is pullback along the Frobenius of C, which is the inverse of this Galois action on $\overline{\mathbf{F}}_q$-points (we compare the two Frobenius maps in Section 12.1*). Hence F acts on $\varprojlim_k H^1(C_{ét}, \mu_{\ell^k})$ by π^{-1}. Weil showed [Wei48b] that the characteristic polynomial of π on $T_\ell J$ has integer coefficients, and that the numerator of the zeta function is

$$P_1(T) = \det(1 - T\pi \mid T_\ell J).$$

Because of the twist, F acts on the untwisted group $H^1(C, \mathbf{Q}_\ell)$ by $q\pi^{-1}$ (Section 10.5*). Its eigenvalues q/α run through the eigenvalues α of π again, by the functional equation, so also $P_1(T) = \det(1 - TF \mid H^1(C, \mathbf{Q}_\ell))$, as predicted in Section 1.1. For the curve C_0 above, π acts on $T_\ell J$ with characteristic polynomial $T^2 + 3$, and indeed $P_1(T) = 1 + 3T^2$.

The example suggests how to obtain a theory with coefficients of characteristic zero: compute étale cohomology with the finite coefficients $\mathbf{Z}/\ell^k$, where the theory works, take the inverse limit over k to obtain $\mathbf{Z}_\ell$-modules, and finally invert ℓ. The limit has to be taken with care, as $\mathrm{Spec}\,\mathbf{F}_q$ already shows. On it, étale sheaves are sets with a continuous action of the Galois group $\widehat{\mathbf{Z}}$, and H^1 with coefficients in an abelian group A is the group of continuous homomorphisms $\widehat{\mathbf{Z}} \to A$. For $A = \mathbf{Z}/\ell^k$ this is $\mathbf{Z}/\ell^k$, and the limit of these groups is $\mathbf{Z}_\ell$. But the étale topology sees $\mathbf{Z}_\ell$

only as a discrete group, and a continuous homomorphism from the compact group $\widehat{\mathbf{Z}}$ to the discrete torsion-free group $\mathbf{Z}_\ell$ is zero, so the constant sheaf $\mathbf{Z}_\ell$ has $H^1 = 0$. What one wants, therefore, is the limit of the cohomology groups, not the cohomology of the limit sheaf. After inverting ℓ this gives a theory with coefficients in $\mathbf{Q}_\ell$. However, the eigenvalues of Frobenius lie in an algebraic closure $\overline{\mathbf{Q}}_\ell$ of $\mathbf{Q}_\ell$ rather than in $\mathbf{Q}_\ell$ itself, and so we also allow coefficients in $\overline{\mathbf{Q}}_\ell$, where characteristic polynomials split.

The resulting theory is ℓ-adic cohomology, for a prime ℓ not dividing q. It was constructed by Grothendieck and his collaborators, who proved (a), (b) and (c) for it in SGA 4 and SGA 5 [SGA4], [SGA5]. Purity (d) is the hard part, and Deligne proved it [Del74], [Del80]. The general form of his theorem, on weights of $Rf_!$, was later reproved by Laumon with the ℓ-adic Fourier transform [Lau87] (see also [Kat88], [KW01]); we call it the Deligne–Laumon theorem. Our proof uses the Fourier transform as Laumon does, but it is organized differently in places.

Of statement (4) we prove that $b_i = b_{2d-i}$. We also prove that $b_i = \dim H^i(X, \mathbf{Q}_\ell)$ for every ℓ not dividing q; in particular this dimension does not depend on ℓ (Theorem 19.7*). The comparison with the Betti numbers of a lift to characteristic zero needs the comparison theorem between étale and singular cohomology, which we do not prove.

1.3 The function–sheaf dictionary

The proof of purity does not treat one variety at a time. It cuts a variety into pieces and fibres it over other varieties, and point counts behave simply under both operations. If $Z_0 \subset X_0$ is closed with complement U_0, then $\#X_0(\mathbf{F}_q) = \#U_0(\mathbf{F}_q) + \#Z_0(\mathbf{F}_q)$, and for a morphism $f : X_0 \to Y_0$,

$$\#X_0(\mathbf{F}_q) = \sum_{y \in Y_0(\mathbf{F}_q)} \#f^{-1}(y)(\mathbf{F}_q).$$

On the right we sum a function on Y_0 that is not constant. To follow such sums with cohomology, we replace functions by sheaves.

The sheaves in question are *constructible*: there is a decomposition of X into finitely many locally closed pieces on each of which the sheaf is locally constant with finite stalks. After passing to the limit, the stalks are finite-dimensional $\mathbf{Q}_\ell$-vector spaces instead. A *Frobenius structure* on a constructible sheaf K on X is an isomorphism $F^*K \simeq K$, and a sheaf with a Frobenius structure is a *Weil sheaf*. The same definitions apply to complexes of sheaves (Section 10.5*). Let x be a point of X_0 over $\mathbf{F}_{q^m}$. The corresponding point $\bar{x}$ of X is fixed by F^m, so the stalk of $(F^m)^*K$ at $\bar{x}$ is $K_{\bar{x}}$. Iterating the Frobenius structure gives an isomorphism $(F^m)^*K \simeq K$, and on the stalk at $\bar{x}$ it becomes an endomorphism Frob_x of $K_{\bar{x}}$. Its trace defines the *trace function*

$$t_K : X_0(\mathbf{F}_{q^m}) \longrightarrow \mathbf{Q}_\ell, \qquad x \longmapsto \mathrm{Tr}(\mathrm{Frob}_x \mid K_{\bar{x}}),$$

where for a complex the trace is the alternating sum of the traces on the cohomology groups. The constant sheaf has trace function 1.

Operations on functions correspond to operations on sheaves. For example, pulling back a function along f corresponds to the inverse image f^*K, and the product of two functions to the tensor product. Similarly, summing a function over the fibres of f corresponds to the direct image with compact supports $Rf_!K$, whose stalk at a point $\bar{y}$ is the compactly supported cohomology of K on the fibre over $\bar{y}$. The resulting identity $t_{Rf_!K}(y) = \sum_{f(x)=y} t_K(x)$ is a form of the trace formula (Corollary 12.26*). When f is the map to a point, it is (b) with coefficients in K. The one further operation we need is a Fourier transform, explained in Section 1.6. This correspondence between functions and sheaves is Grothendieck's *function–sheaf dictionary* [SGA4½, Sommes trig.]:

functions on $X_0(\mathbf{F}_{q^m})$	Weil complexes on X
$t \mapsto t \circ g$	g^*K
$t \cdot t'$	$K \otimes^{\mathbf{L}} L$
$t \mapsto \big(y \mapsto \sum_{f(x)=y} t(x)\big)$	$Rf_!K$
$t = t\vert_U + t\vert_Z$	$j_!j^*K \to K \to i_*i^*K \to$
$\sum_x t(x)$	$R\Gamma_c(X,K)$ with its Frobenius
$t \mapsto \big(y \mapsto -\sum_x t(x)\psi(\mathrm{Tr}_{\mathbf{F}_{q^m}/\mathbf{F}_q}(xy))\big)$	$Rp_{2!}(p_1^*K \otimes \mathcal{K}_\psi)[1]$

Proposition 12.28* states and proves the first five lines. The last line is built from the first three, as we explain in Section 1.6. Since $Rf_!$ of a sheaf is a complex, we work in derived categories throughout.

1.4 The six operations

Duality has no simple counterpart for functions, and it needs two more operations: the internal Hom of sheaves and the *exceptional inverse image* $f^!$, a right adjoint of $Rf_!$. With f^*, Rf_*, $Rf_!$ and $\otimes$ these are Grothendieck's *six operations*. Part I of these notes, roughly their first third, constructs them for all the coefficient rings we need. To see why $f^!$ is the right tool, consider Poincaré duality for a smooth variety X of pure dimension d over an algebraically closed field k, with coefficients $\Lambda = \mathbf{Z}/\ell^n$. It should say that cup product

$$H^i_c(X,\Lambda) \times H^{2d-i}(X,\Lambda(d)) \longrightarrow H^{2d}_c(X,\Lambda(d)) \longrightarrow \Lambda$$

is a perfect pairing. Here $\Lambda(d) = \mu_{\ell^n}^{\otimes d}$ is the *Tate twist* (Definition 2.53*). The twist is needed because Frobenius does not act trivially on top cohomology. For instance, the map $F : \mathbf{P}^1 \to \mathbf{P}^1$ has degree q. It pulls the class of a point back to q times itself, so F acts on $H^2(\mathbf{P}^1,\Lambda)$ by q. With our conventions F acts on $\Lambda(1)$ by q^{-1} (Section 2.12*), and therefore it acts trivially on $H^2(\mathbf{P}^1,\Lambda(1))$. In general the twist (d) removes the factor q^d from the pairing.

For an oriented manifold V of dimension r there is a clean proof, due to Verdier. Let $a : V \to \mathrm{pt}$ be the map to a point. The functor $Ra_!$, cohomology with compact supports, has a right adjoint $a^!$, and the adjunction gives, for every sheaf G on V,

$$R\mathrm{Hom}\big(R\Gamma_c(V,G),\Lambda\big) \simeq R\mathrm{Hom}(G, a^!\Lambda).$$

This isomorphism comes from the adjunction alone and holds for every V; what makes it a duality theorem is a computation of $a^!\Lambda$. It is a complex of sheaves on V, and its restriction to an open subset U is the corresponding complex for U. So it can be computed in a chart $\mathbf{R}^r$. There its only cohomology sheaf sits in degree $-r$, with stalks $\mathrm{Hom}(H^r_c(\mathbf{R}^r,\Lambda),\Lambda) \simeq \Lambda$. An orientation identifies these stalks consistently, and this gives $a^!\Lambda \simeq \Lambda[r]$. For $G = \Lambda$ the displayed isomorphism then reads $R\mathrm{Hom}(R\Gamma_c(V,\Lambda),\Lambda) \simeq R\mathrm{Hom}(\Lambda,\Lambda[r]) = R\Gamma(V,\Lambda)[r]$. Taking cohomology in degree $-i$ and using that Λ is a field or, for $\Lambda = \mathbf{Z}/\ell^n$, a self-injective ring, this says $\mathrm{Hom}(H^i_c(V,\Lambda),\Lambda) \simeq H^{r-i}(V,\Lambda)$, which is Poincaré duality for V.

We follow the same route. For the structure morphism a of a smooth variety X of dimension d we show $a^!\Lambda \simeq \Lambda_X(d)[2d]$, again by a local computation (*smooth purity*). Poincaré duality follows.

The main theorem of Part I is that $f^!$ exists and carries constructible complexes to constructible complexes (Theorem 5.2*). Existence is a general fact. Indeed, to give a right adjoint of $Rf_!$ is to find, for each N, an object $f^!N$ with $\mathrm{Hom}(M, f^!N) = \mathrm{Hom}(Rf_!M, N)$, naturally in M, and Brown's representability theorem provides such objects for an exact functor between triangulated

categories that commutes with arbitrary direct sums. It needs one hypothesis on the source: it must be *compactly generated*, that is, built by shifts, cones and direct sums from a set of objects M for which $\mathrm{Hom}(M, -)$ commutes with direct sums. Such objects are called compact; in the derived category of a ring, for example, the perfect complexes are the compact objects. Constructible complexes are such objects, but they are not closed under direct sums. We therefore apply the theorem to the larger category built from them in this way.

It remains to show that $f^!$ carries constructible complexes to constructible complexes, and this is geometric. Locally f is a closed immersion followed by a projection from affine space, and affine space is open in a product of projective lines, so two cases remain. For the projection $\mathbf{P}^1_Y \to Y$, we compute $f^!$ explicitly, whereas for a closed immersion i with open complement j, a distinguished triangle computes $i^!K$ from K and Rj_*j^*K. This case therefore comes down to the theorem of Deligne that the ordinary direct image Rj_* preserves constructible complexes.

The six operations must also be available for $\mathbf{Z}_\ell$- and $\mathbf{Q}_\ell$-coefficients, compatibly with reduction modulo ℓ^n. This is a problem, because inverse limits behave badly on the étale site: on $\operatorname{Spec}\mathbf{F}_q$, as we saw in Section 1.2, the constant sheaf $\mathbf{Z}_\ell$ has $H^1 = 0$, although the limit of the groups $H^1(\operatorname{Spec}\mathbf{F}_q, \mathbf{Z}/\ell^k)$ is $\mathbf{Z}_\ell$. For this reason the classical theory works with compatible systems of sheaves modulo ℓ^n and builds the derived categories by hand [Eke90]. We use instead the pro-étale site of Bhatt and Scholze [BS15]. Its objects over X include, besides the étale maps $U \to X$, inverse limits $\varprojlim_i U_i \to X$ of affine étale maps, for example the spectrum of an algebraic closure of $\mathbf{F}_q$ over $\operatorname{Spec}\mathbf{F}_q$, and its coverings are the families that are jointly surjective in a strong sense. Every étale sheaf defines a sheaf on it, and every bounded below complex of étale sheaves has the same cohomology on the two sites (Chapter 2*). The gain is that $\mathbf{Z}_\ell$, with its topology, becomes a sheaf whose cohomology is the limit: on the pro-étale site $H^1(\operatorname{Spec}\mathbf{F}_q, \mathbf{Z}_\ell) = \mathbf{Z}_\ell$ (Example 2.57*). On it, $\mathbf{Z}_\ell$ is a sheaf of rings whose sections are continuous functions to $\mathbf{Z}_\ell$, so that an ℓ-adic sheaf is simply a sheaf of modules over it, and we use the usual derived category of these modules, with no inverse systems. In this setting all the geometric arguments are made with coefficients $\mathbf{Z}/\ell^n$, where the classical theory applies, and the pro-étale site then carries them over to $\mathbf{Z}_\ell$, $\mathbf{Q}_\ell$ and $\overline{\mathbf{Q}}_\ell$.

1.5 The trace formula

Part II proves Poincaré duality and the trace formula. Why should the Lefschetz formula hold for Frobenius? For a self-map φ of a compact manifold V whose fixed points are transverse, the Lefschetz fixed point formula reads

$$\sum_i (-1)^i \operatorname{Tr}(\varphi^* \mid H^i(V, \mathbf{Q})) = \sum_{\varphi(x)=x} \operatorname{sign}\det(1 - d\varphi_x),$$

so each fixed point x is counted with the sign of $\det(1 - d\varphi_x)$. In our case the differential of Frobenius is zero, because $d(u^q) = qu^{q-1}du = 0$ in characteristic p. So every fixed point should count with $+1$. On a curve one can see this directly. Let u be a local coordinate at a fixed point, and v the same coordinate on the second factor of $X \times X$. In these coordinates the graph of F^m is $v = u^{q^m}$, and it meets the diagonal $v = u$ with multiplicity one.

The Grothendieck–Lefschetz trace formula states that for every Weil complex K,

$$\operatorname{Tr}(F^m \mid R\Gamma_c(X, K)) = \sum_{y \in X_0(\mathbf{F}_{q^m})} t_K(y)$$

(Theorem 12.24*). For the constant sheaf it is axiom (b). In Example 12.32* we apply it to the elliptic curve $C_0 \colon y^2 = x^3 - x$ over $\mathbf{F}_3$ of Section 1.1, where it reads $N_m = 1 - \operatorname{Tr}(F^m \mid H^1) + 3^m$.

Since $N_1 = 4$, the trace of F on H^1 is 0. Duality gives $\det(F \mid H^1) = 3$, and we recover $Z(C_0, T) = (1 + 3T^2)/((1 - T)(1 - 3T))$.

We deduce the trace formula, as in SGA 5 [SGA5, III, III B, XV], from a general Lefschetz formula. A *correspondence* on X is a pair of maps $c_1, c_2 : C \to X$. For instance, the graph of a map φ is a correspondence with $c_1 = \mathrm{id}$ and $c_2 = \varphi$. To act on the cohomology of K, a correspondence needs a map $u : c_2^* K \to c_1^! K$, and for the graph of F this is the Frobenius structure $F^* K \to K$. For such data the Lefschetz–Verdier formula (Theorem 11.13*) holds when X and C are proper. It writes the trace on cohomology as a sum, over the fixed points, of *local terms*, numbers computed from K and u near each fixed point. In general, however, a local term is not the trace on the stalk.

To apply this to Frobenius on a curve X, which need not be proper, we compactify X by adding finitely many points and extend K by zero. The added points are again fixed by Frobenius, but there K is no longer locally constant: its stalk is zero, while nearby stalks need not be. The main computation of Part II shows that even so, at every fixed point y of Frobenius on a curve, the local term is the trace of Frob_y on $K_{\bar{y}}$ (Proposition 12.16*). To prove this, we cut K into its restrictions to y and to the complement of y. Local terms are additive for this decomposition, and for the piece supported at y the local term is the trace. It therefore suffices to show that the local term vanishes when $K_{\bar{y}} = 0$.

Here $dF = 0$ is used once more. Because of it, at (y, y) the graph $v = u^{q^m}$ is tangent to the axis $v = 0$, that is, to $X \times \{y\}$, while the diagonal $v = u$ is not. Blowing up the point (y, y) of $X \times X$ therefore separates the two curves. After the blow-up the local term is a map between cohomology groups supported near two disjoint curves meeting the exceptional divisor, and such a map factors through the stalk of K at y; so it vanishes when $K_{\bar{y}} = 0$ (Lemma 12.15*). (In Chapter 12* we use instead the transposed graph $\{(F^m x, x)\}$, which is tangent to $\{y\} \times X$.)

Curves suffice for the general case. Indeed, the trace formula is compatible with $Rf_!$, and a variety can be fibred in curves over a variety of smaller dimension.

1.6 The Deligne–Laumon theorem

Part III proves purity. Fix an embedding $\iota : \overline{\mathbf{Q}}_\ell \to \mathbf{C}$, and write q_x for the number of elements of the residue field of a closed point x of X_0. A Weil sheaf with coefficients in $\overline{\mathbf{Q}}_\ell$ is *pure of weight* w if every eigenvalue α of Frobenius on every stalk satisfies $|\iota(\alpha)| = q_x^{w/2}$, and *mixed* if it has a finite filtration whose quotients are pure. Similarly, a complex has *weights at most* w if each cohomology sheaf $\mathcal{H}^i$ is mixed with weights at most $w + i$ (Definition 13.1*). The shift by i is chosen so that both the constant sheaf, pure of weight 0, and its cohomology complex $R\Gamma(X, \overline{\mathbf{Q}}_\ell)$, whose H^i should have weight i, have weights at most 0.

The *Deligne–Laumon theorem* (Theorem 13.11*), proved by Deligne in Weil II [Del80] and reproved by Laumon with the Fourier transform [Lau87], says that if K has weights at most w, so does $Rf_! K$ for every morphism f of separated $\mathbf{F}_q$-schemes of finite type. It gives only an upper bound. For the constant sheaf on a smooth proper X_0, duality supplies the lower one. With f the map to a point, the theorem says that the eigenvalues on H^i satisfy $|\iota(\alpha)| \le q^{i/2}$. Since Poincaré duality pairs α on H^i with q^d/α on H^{2d-i}, the same bound in degree $2d - i$ gives $|\iota(\alpha)| \ge q^{i/2}$. The two bounds together are purity. Theorem 19.7* deduces the Weil conjectures from it.

The proof has four steps, each of which has a model in classical mathematics.

Step 1: from varieties to curves. The theorem is stated for all sheaves, not only the constant one, because the proof is an induction on dimension. Away from a closed subset, a variety is fibred in curves over a variety of smaller dimension, much as one computes the cohomology of a

fibre bundle by the Leray spectral sequence from the cohomology of the fibres and of the base. Along such a fibration, however, the direct image of the constant sheaf is no longer constant, so the induction needs the theorem for all sheaves. It thereby reduces to curves, provided the direct images that occur along the way are mixed (Theorem 18.8*). On a curve the statement says: if $\mathcal{F}$ is a lisse sheaf, pure of weight w, on a smooth curve U_0, then $H^1_c(U, \mathcal{F})$ has weights at most $w+1$ (Theorem 17.15*). Since every curve, after removing finitely many points, maps to the affine line by a finite map, it is enough to treat open subsets of $\mathbf{A}^1$. The proof on curves combines two ideas: Rankin's positivity argument, which Deligne used in Weil I and Weil II [Del74], [Del80], and the Fourier transform, which Laumon [Lau87] used in place of Deligne's monodromy arguments.

Step 2: positivity bounds the eigenvalues at points. The model is Rankin's method for the coefficients of modular forms [Ran39]: the Dirichlet series $\sum |\tau(n)|^2 n^{-s}$ has nonnegative coefficients, so the position of its first pole bounds each individual coefficient. Here the role of the Dirichlet series is played by an L-function. Let $\mathcal{G}$ be a *lisse* sheaf on an affine curve U_0, that is, a locally constant one. Such a sheaf is the same as a representation of the fundamental group, and its *monodromy* is the image of that representation. Call $\mathcal{G}$ *real* if the characteristic polynomials of Frobenius on its stalks have real coefficients (Definition 14.21*). Its L-function is the product, over closed points x, of the *Euler factors* $\det(1 - \mathrm{Frob}_x\, T^{\deg x} \mid \mathcal{G}_{\bar{x}})^{-1}$. On an affine curve $H^0_c = 0$ and $H^i_c = 0$ for $i > 2$, so the trace formula shows that the L-function equals $\det(1 - TF \mid H^1_c)/\det(1 - TF \mid H^2_c)$. Two facts make this useful.

- Consider an even tensor power $\mathcal{G}^{\otimes 2m}$. The logarithm of an Euler factor has coefficients $\mathrm{Tr}(\mathrm{Frob}^n_x \mid \mathcal{G}_{\bar{x}})^{2m}/n$, and these are nonnegative because the traces are real. Each Euler factor therefore has nonnegative coefficients bounded by those of the whole product. In particular it converges wherever the product does (Lemma 14.24*).
- The product has poles only where $\det(1 - TF \mid H^2_c)$ vanishes, and H^2_c is easy to compute: it is the largest quotient of a stalk on which the monodromy acts trivially, twisted by (-1).

The second fact bounds the eigenvalues on H^2_c of $\mathcal{G}^{\otimes 2m}$, and the first turns this into a bound at every point. To state the result, let β be the largest *determinant weight* of the irreducible constituents of $\mathcal{G}$; the determinant weight of a constituent of rank r is the weight of its determinant divided by r (Definition 14.7*). The eigenvalues on H^2_c of $\mathcal{G}^{\otimes 2m}$ have absolute value at most $q^{m\beta+1}$. Indeed, by the second fact H^2_c is a quotient of a stalk of $\mathcal{G}^{\otimes 2m}$ on which the geometric monodromy acts trivially, twisted by (-1). On such a quotient Frobenius acts through one-dimensional pieces, each a constituent of $\mathcal{G}^{\otimes 2m}$, and the determinant weight of every constituent of $\mathcal{G}^{\otimes 2m}$ is at most $2m\beta$ (Proposition 14.19*). So these eigenvalues have absolute value at most $q^{m\beta}$, and the twist (-1) multiplies them by q. The product therefore converges for $|T| < q^{-m\beta-1}$, and by the first fact the Euler factor at x converges there too. This factor has a pole where $T^{\deg x} = \alpha^{-2m}$, for each eigenvalue α of Frob_x on $\mathcal{G}_{\bar{x}}$. Hence $|\iota(\alpha)|^{2m} \le q_x^{m\beta+1}$ for every m, and taking $2m$-th roots and letting $m \to \infty$ gives the intermediate result:

every Frobenius eigenvalue α at a point of a real lisse sheaf satisfies $|\iota(\alpha)|^2 \le q_x^{\beta}$

(Proposition 14.25*).

Step 3: rank one, and purity of real sheaves. The bound of Step 2 is only useful if the determinants have weights, that is, if lisse sheaves of rank one are pure. A rank-one sheaf is a character of the fundamental group, the analogue of a Hecke character, and it is pure as soon as its restriction to the geometric fundamental group has finite image. We show that this image is finite. If it were not, the logarithm of a power of the character would give a nonzero class in H^1 of the curve fixed by Frobenius. This class cannot come from the removed points, since Frobenius acts on those classes by powers of q. So it comes from the compactification, and it gives a nonzero

vector in the Tate module of its Jacobian fixed by Frobenius. Such a vector is a compatible system of rational ℓ-power torsion points of unbounded order, which is impossible because the Jacobian has only finitely many rational points (Section 14.1*). With the determinants under control, Step 2 bounds the eigenvalues of $\mathcal{G}$ by its largest determinant weight. To bound those of a single constituent we apply Step 2 to exterior powers of $\mathcal{G}$, in which that constituent is multiplied by the determinants of the others, which are now known; this isolates its weight (as in the example $\overline{\mathbf{Q}}_\ell \oplus \overline{\mathbf{Q}}_\ell(-1)$ of Chapter 14*). Applied also to the dual, which gives the lower bound, it yields the second intermediate result:

each irreducible constituent of a real lisse sheaf on a curve is pure

(Theorem 14.26*).

Step 4: the Fourier transform turns cohomology into a stalk. Steps 2 and 3 bound eigenvalues at the points of a curve, but we must bound eigenvalues on its cohomology. The Fourier transform turns one into the other. Its model is the discrete Fourier transform $\hat{t}(y) = \sum_{x\in\mathbf{F}_q} t(x)\psi(xy)$ of a function t on $\mathbf{F}_q$, where ψ is a nontrivial additive character. Two properties matter. The value $\hat{t}(0)$ is the sum of all values of t, so a sum over the line becomes a single value. And Plancherel's formula $\sum_y |\hat{t}(y)|^2 = q\sum_x |t(x)|^2$ holds.

The classical example is the Gauss sum. Let χ be a nontrivial multiplicative character, extended by $\chi(0) = 0$, and let $g = \sum_x \chi(x)\psi(x)$ be its Gauss sum. Substituting $x \mapsto x/y$ shows that $\hat{\chi}(y) = \overline{\chi}(y)g$ for $y \neq 0$, so the transform has the same absolute value $|g|$ at all $q-1$ nonzero points, and $\hat{\chi}(0) = 0$. Plancherel now reads $(q-1)|g|^2 = q(q-1)$, and we find $|g| = \sqrt{q}$. In other words, a function of absolute value 1 is carried to one of absolute value $\sqrt{q}$, that is, weight 0 to weight 1. This is the shape of the whole argument: first the transform is shown to have a single absolute value (to be pure), and then Plancherel finds which one.

Grothendieck's dictionary turns the transform into an operation on sheaves on the affine line. Read the formula for $\hat{t}$ from the inside out: pull t back to the plane along $p_1(x,y) = x$, multiply by the function $\psi(xy)$, and sum over the fibres of $p_2(x,y) = y$. Each step is a line of the table in Section 1.3. So if $\mathcal{K}_\psi$ is a sheaf on the plane with trace function $\psi(xy)$ and we put

$$\mathrm{FT}_\psi K = Rp_{2!}(p_1^* K \otimes \mathcal{K}_\psi)[1],$$

then for $y \in \mathbf{F}_q$ the table gives

$$t_{\mathrm{FT}_\psi K}(y) = -\,t_{Rp_{2!}(p_1^*K\otimes\mathcal{K}_\psi)}(y) = -\sum_{x\in\mathbf{F}_q} t_{p_1^*K}(x,y)\, t_{\mathcal{K}_\psi}(x,y) = -\sum_{x\in\mathbf{F}_q} t_K(x)\,\psi(xy) = -\,\hat{t}_K(y).$$

The sign comes from the shift [1], which changes the sign of every trace.

Now let M be an irreducible lisse sheaf, pure of weight w, on a dense open $j : U_0 \to \mathbf{A}^1$, and let j_*M be its direct image to $\mathbf{A}^1$. Its compactly supported cohomology lives in degrees 1 and 2; as in Step 2, H^2_c is easy, and the group we must bound is $H^1_c(\mathbf{A}^1, j_*M)$. Just as $\hat{t}(0)$ is the sum of t, the stalk at 0 of the Fourier transform of j_*M is the compactly supported cohomology of j_*M. The sheaves $\mathcal{L}_{\psi(ax)}$ with trace function $x \mapsto \psi(ax)$ are exceptions, as the characters $\psi(ax)$ are for the discrete transform: their transform is concentrated at one point. Suppose that no constituent of M over $\overline{\mathbf{F}}_q$ is of this form. Then the transform is again of the form j'_*M' for a lisse sheaf M' on a dense open of the dual line (Proposition 16.17*). The hypothesis excludes in particular constant constituents (the case $a = 0$), so the monodromy of M has no coinvariants and $H^2_c(\mathbf{A}^1, j_*M) = 0$. This gives the third intermediate result:

the stalk of j'_*M' at 0 is exactly $H^1_c(\mathbf{A}^1, j_*M)$.

It remains to find the weight of M', and here the two halves of the Gauss sum argument return. To show that M' is pure we use Step 3: we twist M to weight 0 and add its dual, whose eigenvalues are the complex conjugates. The transform of the result is again a real sheaf, and it maps onto a twist of M', which is irreducible. So a twist of M' is an irreducible constituent of a real sheaf, hence pure by Step 3, and therefore M' is pure (Proposition 17.4*). The weight of M' is then found from the sheaf version of Plancherel's formula, which compares the sums of $|t_M(x)|^2$ and $|t_{M'}(y)|^2$ over $\mathbf{F}_{q^m}$ as m grows: it is $w+1$ (Theorem 17.5*). By Proposition 15.5*, the stalk of j'_*M' at 0 has weights at most $w+1$, and this is the bound we want. For genus-one curves over $\mathbf{F}_3$, the resulting purity gives $1 \le N_1 \le 7$ (Example 19.9*).

1.7 Outline

The main theorem is the Weil conjectures for smooth proper varieties over $\mathbf{F}_q$ (Theorem 19.7*), with the statement on Betti numbers as above. The proof follows Weil's plan: it constructs a cohomology theory satisfying the properties (a)–(d) of a Weil cohomology listed in Section 1.2. Each part uses the one before it.

- Part I (Chapters 2*–Chapter 8*) constructs the six operations for ℓ-adic coefficients (Theorem 5.2*). It ends with smooth base change and the Künneth formula.
- Part II (Chapters 9*–Chapter 12*) proves smooth purity (Theorem 9.3*), Verdier and Poincaré duality (Theorem 10.10*), which is (c), and the trace formula, which is (b). It ends with rationality and the functional equation. As for (a), the groups H^i_c are finite because $Rf_!$ preserves constructibility. They vanish outside $[0, 2\dim X]$, and Theorem 10.15* treats the smooth proper case used for the Weil conjectures.
- Part III (Chapters 13*–Chapter 19*) defines weights, proves purity of real sheaves on curves, introduces the Fourier transform and proves the Deligne–Laumon theorem, first on curves and then in general. It ends with the Weil conjectures.

Most appendices prove results from algebraic geometry that the main text uses: Nagata compactification and Chow's lemma, abelian varieties, the Picard scheme of a curve, the cohomology of curves with torsion coefficients, and facts on algebraic groups. Appendix E* is of a different kind. It is not a preliminary but a second, independent theory of local terms, by specialization to the normal cone, which computes the local terms of contracting correspondences; the main line of the proof does not use it.

1.8 Prerequisites and reading guide

We assume a first course in algebraic geometry: schemes, quasi-coherent sheaves and their cohomology, curves and divisors, and étale morphisms; some homological algebra (derived functors, derived and triangulated categories); and singular cohomology with Poincaré duality and the Lefschetz fixed point formula, which serve as models. We do not assume étale cohomology. The results from the étale theory with torsion coefficients that we need [SGA4], [SGA4½] are stated where they are used, with references.

On a first reading one can take the foundations on trust and follow the main line:

- the statements of Chapter 3* and of the main theorem of Part I, Theorem 5.2*;
- the base change map and the statement of proper base change (Theorem 4.33*);
- Chapter 6* on why $f^!$ preserves constructibility with torsion coefficients;

- smooth purity with finite coefficients (Section 9.2*), which is where the twist (d), the shift $[2d]$ and the trace map come from;
- Chapter 10* on duality;
- Frobenius and Weil complexes (Section 10.5*); in Chapter 12*, the local calculation at a fixed point of Frobenius on a curve;
- Chapters 13*–Chapter 19*, beginning with the plan in the opening of Part III.

The passages between coefficient rings (most of which are applications of Proposition 7.12*), the coherence of the various comparison maps, smooth base change (Chapter 8*), and the pro-étale foundations of Chapter 2* can be read when needed.

1.9 Conventions

We fix a prime ℓ and a field k that is finite or algebraically closed, of characteristic different from ℓ. Unless said otherwise, schemes are separated and of finite type over k, and morphisms are k-morphisms. The hypothesis on k is needed to bound the ℓ-cohomological dimension of k (Lemma 2.46*), while the constructions of Sections 2.12* and Section 3.1* make sense for arbitrary schemes. Finally, a citation such as [Stacks, 0F0C] refers to the tag of a result in the Stacks Project [Stacks].

On the pro-étale site we regard every coefficient ring Λ as a condensed ring, that is, as a sheaf of rings on profinite sets, which remembers the topology of Λ. (On the étale site, which we use only with finite coefficients, a finite ring is simply a ring.) The coefficient sheaf Λ_X on X is then the pullback of Λ along the map from X to a point (Section 2.12*). For a finite ring A this pullback is the classical constant sheaf.

Derived categories are the classical ones. Namely, $D(X_{\text{pro\'et}}, \Lambda)$ is the derived category of the abelian category of sheaves of Λ_X-modules. On these categories all functors are derived, and we write them f^*, Rf_*, $Rf_!$ and $f^!$. Unless another category is named, we have $f^* : D(Y_{\text{pro\'et}}, \Lambda) \to D(X_{\text{pro\'et}}, \Lambda)$ and $Rf_* : D(X_{\text{pro\'et}}, \Lambda) \to D(Y_{\text{pro\'et}}, \Lambda)$, whereas the categories on which $Rf_!$ and $f^!$ act are fixed where these functors are defined. To avoid set-theoretic problems, we take sizes with respect to a fixed universe that is large enough to contain the w-contractible covers of Theorem 2.13*.

The table below lists the main notation, together with the technical terms (lisse, Weil structure, pure, mixed, real) whose meaning in these notes is fixed by a definition, and the place where each is defined. The Frobenius F of the Introduction, which raises coordinates to their q-th powers, is written Φ_X in the body of the notes. A Frobenius structure $F^*K \simeq K$ becomes there a Weil structure $\mathrm{Fr}_X^* K \simeq K$ for the geometric Frobenius Fr_X, and Lemma 12.3* shows that the two give the same operators.

Notation

symbol	meaning	defined in
$X_{\mathrm{pro\acute{e}t}}$	the pro-étale site	Definition 2.4*
Λ_X, $\Lambda_X(d)$	coefficient sheaf, Tate twists	Definition 2.53*
$D_c^b(X,\Lambda)$, $D_{\mathrm{ctf}}^b(X,\Lambda)$	constructible complexes	Definition 3.1*
lisse	locally of constant finite presentation	Definition 3.2*
K_n	reduction modulo π_E^n	Definition 2.61*
cd_ℓ	cohomological dimension	Definition 2.45*
$R\underline{\Gamma}_Z$	local cohomology	Definition 2.59*
$\mathrm{Loc}(D_{\mathrm{ctf}}^b(X,\Lambda))$	localizing subcategory	Definition 5.1*
$Rf_!$, $f^!$	exceptional operations	Theorem 5.2*
ω_f	relative dualizing complex	Definition 9.4*
Tr_f	trace map	Definition 9.24*
$\mathbb{D}_X$	Verdier dual	Definition 10.2*
Fr_X, Weil structure	geometric Frobenius, $\mathrm{Fr}_X^* K \simeq K$	Definition 10.18*
Φ_X	q-power Frobenius	Definition 12.2*
c, $\mathrm{Fix}(c)$	correspondences	Definition 11.5*
$T_c(u)$	action of a correspondence	Definition 11.11*
$\mathrm{Tr}_c(u)$, LT	trace class, local terms	Definition 11.9*
pure, mixed	ι-weights	Definition 13.1*
dw_ι	determinant weight	Definition 14.7*
real	real Weil sheaves	Definition 14.21*
$j_*\mathcal{F}$	middle extension	Definition 15.1*
FT_ψ	Fourier transform	Chapter 16*
$T_\ell A$, $V_\ell A$	Tate module	Definition B.13*

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References marked * are to parts of the blueprint this document leaves out:

Chapter 2 The pro-étale site, in I The six operations (`chap-The-pro-etale-site`)

Definition 2.4 The small pro-étale site, in I The six operations (`definition-small-pro-etale-site`)

Theorem 2.13 Existence of contractible covers, in I The six operations (`theorem-existence-contractible-covers`)

Definition 2.45 Cohomological dimension, in I The six operations (`definition-cohomological-dimension`)

Lemma 2.46 Cohomological dimension of the base field, in I The six operations (`lemma-cohomological-dimension-base-field`)

Section 2.12 Coefficient rings, in I The six operations (`sec-Coefficient-rings`)

Definition 2.53 Tate twists, in I The six operations (`definition-tate-twists`)

Example 2.57 The first cohomology of a finite field, in I The six operations (`example-first-cohomology-finite-field`)

Definition 2.59 Local cohomology, in I The six operations (`definition-local-cohomology`)

Definition 2.61 Derived adic completeness, in I The six operations (`definition-derived-adic-completeness`)

Chapter 3 Constructible complexes, in I The six operations (`chap-Constructible-complexes`)

Section 3.1 Derived functors, in I The six operations (`sec-Derived-functors`)

Definition 3.1 Constructible sheaves, in I The six operations (`definition-constructible-sheaves`)

Definition 3.2 Lisse sheaves, in I The six operations (`definition-lisse-sheaves`)

Theorem 4.33 Pro-étale proper base change, in I The six operations (`theorem-proper-base-change-classical-pro`)

Definition 5.1 The localizing subcategory generated by $D^b_{ctf}(X, \Lambda)$, in I The six operations (`definition-localizing-subcategory`)

Theorem 5.2 The exceptional inverse image, in I The six operations (`theorem-exceptional-inverse-image`)

Chapter 6 Constructibility of $f^!$, in I The six operations (`chap-Constructibility-of-f`)

Proposition 7.12 Isomorphisms reduce to finite coefficients, in I The six operations (`proposition-isomorphisms-reduce-finite-coefficients`)

Chapter 8 Smooth base change, in I The six operations (`chap-Smooth-base-change`)

Chapter 9 Smooth purity, in II Duality and the trace formula (`chap-Smooth-purity`)

Theorem 9.3 Smooth purity for all admissible coefficients, in II Duality and the trace formula (`theorem-smooth-purity-all-admissible-coefficients`)

Definition 9.4 Relative dualizing complex, in II Duality and the trace formula (`definition-relative-dualizing-complex`)

Section 9.2 The tensor product map, in II Duality and the trace formula (`sec-The-tensor-product-map`)

Definition 9.24 Trace map, in II Duality and the trace formula (`definition-trace-map`)

Chapter 10 Verdier and Poincaré duality, in II Duality and the trace formula (`chap-Verdier-and-Poincare-duality`)

Definition 10.2 Verdier dual and evaluation, in II Duality and the trace formula (`definition-verdier-dual-evaluation`)

Theorem 10.10 Global Verdier duality, in II Duality and the trace formula (`theorem-global-verdier-duality`)

Theorem 10.15 The trace isomorphism in top degree, in II Duality and the trace formula (theorem-trace-isomorphism-top-degree)

Section 10.5 Weil complexes, in II Duality and the trace formula (sec-Weil-complexes)

Definition 10.18 Weil structure, in II Duality and the trace formula (definition-weil-structure)

Example 10.23 The running example: duality, in II Duality and the trace formula (example-running-example-duality)

Definition 11.5 Correspondences, in II Duality and the trace formula (definition-correspondence)

Definition 11.9 Trace class and local terms, in II Duality and the trace formula (definition-trace-class-local-terms)

Definition 11.11 Action of a correspondence on cohomology, in II Duality and the trace formula (definition-action-correspondence)

Theorem 11.13 Proper pushforward and Lefschetz–Verdier, in II Duality and the trace formula (theorem-proper-pushforward-lefschetz-verdier)

Chapter 12 The trace formula, in II Duality and the trace formula (chap-The-trace-formula)

Section 12.1 Frobenius correspondences, in II Duality and the trace formula (sec-Frobenius-correspondences)

Definition 12.2 Frobenius correspondences and their operators, in II Duality and the trace formula (definition-frobenius-correspondences-their-operators)

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